Course Title: Research Methodology in Computer Application (3 Cr.)
Course Code: MCA505
Year/Semester: I/I
Class Load: 5 Hrs. / Week (Theory: 3 Hrs. Practical: 2 Hrs.)

Course Description:

This course is about understanding the importance of research in computer application in terms of human computer interaction. It provides the basic foundation of research in computer application and the process of entire research in computer application which is discussed under research methodology in computer application. The course begins with the introduction and importance of research in computer application and includes the essential components of research process such as philosophical assumptions, conceptualization, literature review, identifying research gap and formulating research problem/questions/objectives/hypothesis, design the research focusing on measurement, instrumentation and data collection. The focus of the course is also computer/software based data analysis and interpretation techniques and skills which produced scientific knowledge Along with theoretical knowledge the course emphasizes on developing practical skills among students so that they can easily carryout research in the area of computer application. In order to develop practical skills, based on hands on practice, the course introduces software like, SPSS, Stata, Python, R and so on and engages students in a particular software (based on relevancy and availability of necessary infrastructure including teaching faculty. Finally, the course includes introduction and practice of Statistical software, Plagiarism checker, citation and referencing tools/packages which enable students to be confirmed that the scientific paper they have written avoids plagiarism, appropriately formatted and referenced.

Course Objectives:

The objective of the course is to enable students to plan their research in any topic/issue/subject in the area of computer application and develop knowledge and skill among them based on hands on practice with the help of recent methods and software based teaching. Final, students will be able to conduct their research work themselves and prepare scientific report based on or with the help of computer/statistical software. The specific objectives of the course are:

- To familiarize students with the concept and process of research and research methodology.
- To enable students to describe the process of scientific research and plan/design research.
- To develop practical skills of doing research in the areas of computer application.
- To enable the students to prepare scientific paper in the standard scientific format of the discipline computer application.

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Course Contents:

Unit 1: Introduction to Research Methodology **[4 Hr.]**

Science, social science and computer application, Research and scientific research, Relationship between research, science and computer application, Research methods and methodology.

Required readings:

Lazar, Jonathan, Feng, Jinjuan Heidi, & Hochheiser, Harry. (2017). *Research Methods in Human-Computer Interaction.* Second Edition. Chapter 1, pp. 1-24. Elsevier, Morgan Kaufmann Publishers.

Unit 2: Research Topic, Literature Survey and Problem Statement **[6 Hr.]**

Identifying research topic/issue/theme, Literature survey and preparing bibliography, Doing literature review and identifying research gap, Formulating and stating research problem, Research problem/questions/objectives and hypothesis.

Required readings:

Lazar, Jonathan, Feng, Jinjuan Heidi, & Hochheiser, Harry. (2017). *Research Methods in Human-Computer Interaction.* Second Edition. Chapter 1, pp. 1-24. Elsevier, Morgan Kaufmann Publishers.

Unit 3: Research Design and Instrumentation **[10 Hr.]**

Research Design, Type of research and research design, Descriptive Research, Relational Research and Experimental Research, Relationship Between Descriptive Research, Relational Research, and Experimental Research, Measuring the human, Preparing research design: universe, population and sample size, Sample selection and instrument design, Basics of experimental research, Experimental design, Online research design, Use of qualitative methods in CA: diaries, case studies, interviews and focus groups, Ethnography: interviewing, observing, analyzing, repeating, and theorizing, Automated data collection methods.

Required readings:

Lazar, Jonathan, Feng, Jinjuan Heidi, & Hochheiser, Harry. (2017). *Research Methods in Human-Computer Interaction.* Second Edition. Chapter 2 & 3, pp. 25-69; Chapter 5, pp. 105-133; Chapter 12 & 13, pp. 329-409. Elsevier, Morgan Kaufmann Publishers.

Unit 4: Data Entry, Analysis and Interpretation **[15 Hr.]**

Data entry using software, Preparing data for statistical analysis, Application of statistics in data analysis: descriptive and inferential, Univariate, bivariate and multivariate analysis, Chi-square test, Gamma test and Spearman's Rank Order Correlation, t-test: one sample, independent and paired sample t-tests, F-test: one way and two way ANOVA, Correlation and regression, Non-parametric statistical tests, Machine learning approaches of decision making and prediction, Analyzing qualitative data.

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Required readings:

Lazar, Jonathan, Feng, Jinjuan Heidi, & Hochheiser, Harry. (2017). *Research Methods in Human-Computer Interaction*. Second Edition. Chapter 4, pp. 70-104; Chapter 11, pp. 299-327. Elsevier, Morgan Kaufmann Publishers.

Unit 5: Research Ethics and Writing Research Report **[5 Hr.]**

Working with human subjects, Research report: components and process, Preparing and submitting research proposal and report.

Required readings:

Salmani-Nodoushan, Mohammad Ali, Alavi, Seyyed Mohammad. (2004). *AAP style and research report writing*. Section Three: Reports and Theses, pp. 115-150. Zabankadeh Publications.

Lazar, Jonathan, Feng, Jinjuan Heidi, & Hochheiser, Harry. (2017). *Research Methods in Human-Computer Interaction*. Second Edition. Chapter 15, pp. 454-492. Elsevier, Morgan Kaufmann Publishers.

Wang, Gabe T. & Park, Keumjae. (2016). From Topic Selection to the Complete Paper. Chapter 7, pp. and Chapter 11. Willey Blackwell.

Unit 6: Research based Project Work: **[8 Hr.]**

Selection of a topic, Prepare research proposal, Develop a research plan or Research Design, Collect data and analyze data (computer based), Write a research report.

Required readings:

Salmani-Nodoushan, Mohammad Ali, Alavi, Seyyed Mohammad. (2004). *AAP style and research report writing*. Section Three: Reports and Theses, pp. 115-150. Zabankadeh Publications (info@zabankadeh.net).

Wang, Gabe T. & Park, Keumjae. (2016). *From Topic Selection to the Complete Paper*. Willey Blackwell.

Laboratory Work:

Use of Text editor (LaTex), Data analysis (R Programming/Python), Grammar tool (Grammarly), Referencing tool (Mendeley/EndNote), Plagiarism Checker (Turnitin), Research repository (Google Scholar, Scopus, Zotero),

References:

1. "Research Methodology Methods and Techniques", C. R. Kothari and Gaurav Garg, New Age International, 2019
2. "Research Methodology, A Practical and Scientific Approach" Edited by Vinayak Bairagi and Mousami V. Munot, CRC Press, 2019

3. "Engineering Research Methodology: A Computer Science and Engineering and Information and Communication Technologies Perspective", Krishnan Nallaperumal,

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